



# **AI Engineering Techniques & Architectures Project:** **Meridian Strategy Group** **People & Travel Operations Assistant**

**Business & Product Overview**

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## AI Disclosure

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- Generative AI tools were used during this project as part of the engineering workflow. OpenAI ChatGPT assisted with planning, drafting and review during the project's development.
- The author remains responsible for the project concept, requirements, architecture, implementation decisions, validation, evaluation and submitted content.
- AI-generated outputs informed the design and development process but did not determine the final project decisions.
- Each recommendation was assessed and revised using the project requirements, implementation evidence and observed system behaviour.
- ChatGPT assisted with defining and refining the project concept, use case, business requirements and software requirements. It also helped compare architecture and design options, identify edge cases and evaluation considerations, improve requirements traceability, explain technical concepts such as retrieval-augmented generation (RAG), agent orchestration, Model Context Protocol (MCP), deployment, CI/CD and evaluation, and review project artefacts and documentation.
- AI coding tools assisted with code generation, debugging, testing and repository-level verification. All generated code and technical recommendations were reviewed, tested and adjusted before use.
- The accompanying AI Tooling documentation identifies the tools used, their roles and the relevant AI interactions.

# Executive Summary

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## Company Background and Business Context

Meridian Strategy Group (**MSG**) is a global management consulting organisation comprising both client-facing consulting professionals and enablement functions such as HR, Finance, IT, Legal/Risk and Operations. Client-facing employees regularly work across client engagements and locations, creating a highly mobile workforce whose time is directly connected to client delivery and revenue generation.

While on assignment, employees still need to resolve routine administrative matters such as leave, travel changes, expenses, benefits and approvals. These requests can become surprisingly complex because the correct answer may depend simultaneously on the employee's role, active assignment, travel authorisation, operational records and the applicable company policy or procedure.

For a consulting organisation, the issue is therefore not simply access to HR policies. Employees need to resolve administrative questions quickly, accurately and in the context of their current work, without unnecessarily interrupting client delivery or relying on multiple support teams and systems.

## Agent Overview

Meridian Compass is a fictional internal employee assistant developed for Meridian Strategy Group (MSG). It supports employees whose administrative questions may depend on several factors at once, including their identity, current assignment or travel context, approved company policy, relevant procedures and the appropriate approval relationship.

The challenge is therefore not simply finding a policy document. Employees must also understand how the relevant rule applies to their current circumstances and determine the correct next step. Meridian Compass addresses this need by combining approved internal guidance with authorised operational information to provide a clear, employee-specific response supported by cited evidence.

## Product Proposition

Meridian Compass converts general policy questions into context-specific guidance by drawing on four complementary sources of authority:

| Authority                   | Question Answered                          | Implementation                                                       |
|-----------------------------|--------------------------------------------|----------------------------------------------------------------------|
| Enterprise Identity         | Who is asking?                             | Server-authoritative simulated identity                              |
| Structured operational data | What situation exists?                     | Employee, PTO, assignment, travel, expense and benefits data via MCP |
| Approved policy             | What is permitted, required or prohibited? | Closed 12-policy corpus                                              |
| Approved procedure          | How should the employee proceed?           | Closed 7-procedure corpus                                            |

Where approved Meridian documents and authorised structured data do not provide sufficient support, the system returns a bounded response rather than relying on general model knowledge.

## Primary Employee Journeys

| Journey                         | Employee Need                                     | What the assistant combines                                        |
|---------------------------------|---------------------------------------------------|--------------------------------------------------------------------|
| International-assignment PTO    | Can I take these days off while on assignment?    | Identity, PTO balance, assignment, leave policy, approval role     |
| Role-aware travel entitlement   | What travel entitlement applies to me?            | Identity, role, travel authorisation, travel policy                |
| PTO plus personal extension     | Can I extend a business trip for personal travel? | Travel booking, assignment/PTO context, policy, per diem, approval |
| Expense/per-diem reconciliation | What is reimbursable and what process applies?    | Expense facts, per diem, policy and procedure                      |

## Business Value

Reduce the time and effort required to locate and interpret relevant policy, procedure and operational information.

Provide employees with consistent self-service guidance based on a common, approved evidence base.

Clarify approval routes and procedural next steps where these are supported by authoritative information.

Strengthen transparency by presenting citations, source extracts and an architectural record of tool use.

Reduce AI-related risk by preventing general model knowledge from being represented as Meridian policy.

## Product Principles

| Principle     | Meaning                                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| Grounded      | Company guidance is derived from approved MSG evidence.                                                              |
| Context-aware | The authenticated employee's profile and operational circumstances determine which guidance applies.                 |
| Actionable    | Supported responses provide clear and relevant procedural next steps.                                                |
| Privacy-aware | Access to employee information is governed by authenticated identity and authorised relationships.                   |
| Transparent   | Material guidance is supported by clearly identifiable source evidence.                                              |
| Portable      | The application can be reproduced consistently across local, Docker, continuous integration and hosted environments. |

## Current Implementation

- The final implementation comprises a deployed FastAPI application with:
  - a server-rendered employee interface with simulated Enterprise Identity;
  - controlled agent orchestration;
  - GPT-4.1-mini in the validated production configuration;
  - an official Streamable HTTP MCP boundary;
  - retrieval-augmented generation across 19 approved fictional documents;
  - structured synthetic operational data;
  - controlled mock actions; and
  - deterministic claim verification.
- The retrieval index contains 353 chunks generated with all-MiniLM-L6-v2 embeddings and stored in FAISS IndexFlatIP.
- The MCP layer provides 13 tools.
- Controlled evaluation produced a 90% pass rate, with 27 of 30 cases passing, and the final automated test inventory comprises 138 tests.

## Success Definition

Meridian Compass is considered successful when an authenticated employee can ask a supported work-related question and receive a clear response that reflects the relevant context, uses authorised operational information, retrieves applicable policy and procedure evidence, identifies supported approval requirements, provides practical next steps and citations, and avoids unsupported company determinations where evidence is insufficient.

## Scope Boundaries

- Meridian Strategy Group, its policies and all employee records are fictional or synthetic.
- The assistant does not execute real bookings, HR updates or financial transactions.
- Enterprise Identity is simulated for assessment purposes.
- General web knowledge is not treated as an authoritative source of Meridian policy.
- Where authoritative evidence is unavailable, the assistant provides a bounded response rather than inferring an answer.

## Source Alignment

The business framing is aligned with the final Business Requirements Specification, Software Requirements Specification and Solution Design documents maintained as controlled repository inputs. As no separate Business Overview Markdown file is present at final commit c30c203, this overview retains the established business framing from the earlier Business & Product Overview draft and reconciles it with the final controlled documents and implementation evidence.